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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

IV Semester B. E. Examinations Sept/Oct – 2024

Industrial Engineering and Management

CAD/CAM AND ROBOTICS

Time: 03 Hours ***Maximum Marks: 100***

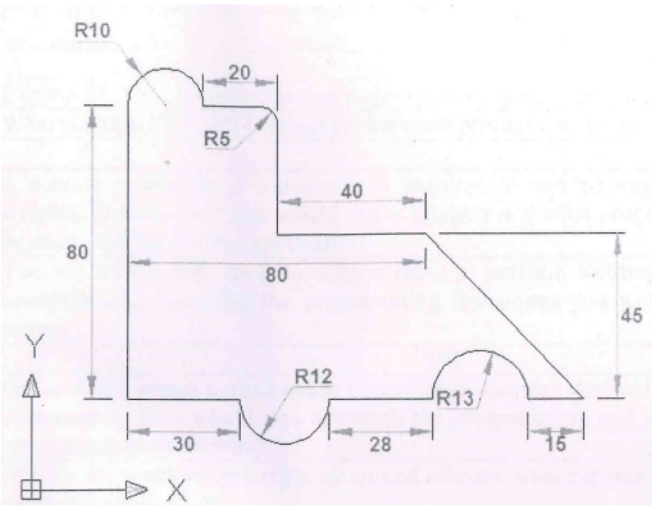
Instructions to candidates:

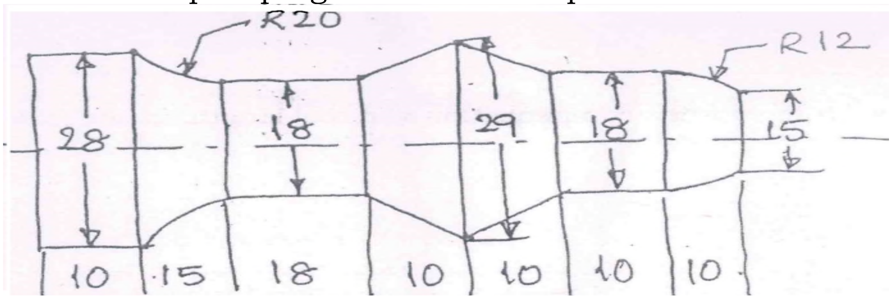
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	What is rendering in computer graphics?	02	2	1
	1.2	What tis the Z-buffer algorithm used for?	02	2	1
	1.3	List the main elements of an NC system.	02	2	2
	1.4	Identify two keys components of a DNC system.	02	1	2
	1.5	How does mechanization differ from automation?	02	1	2
	1.6	What are the primary goals of automation?	02	2	1
	1.7	Define zeroth law of Robotics.	02	1	1
	1.8	List any two features of fixed automation.	02	1	2
	1.9	Define pitch, yaw and roll?	02	2	2
	1.10	What is meant by work envelop?	02	1	1

PART-B

			M	BT	CO
2	a	Consider a scenario where you are designing a complex mechanical part using CAD. Discuss how you would utilize the different stages of the Shigley Model to ensure a successful design.	08	2	1
	b	Express a line from (5,5) to (13,9) on a raster screen using Bresenham's Line Drawing Algorithm. Show the result in a Cartesian graph.	08	3	2
3	a	Figure 3a below shows the final profile to be generated on a MS by using a CNC milling center. Write CNC program and show the tool path.  Fig 3a	12	3	3

	b	What key factors should be considered during the Problem? Definition stage of the Shigley Model when designing a mechanical part?	04	2	2
		OR			
4	a	Explain the APT statements: (i) GOTO and GODLTA (ii) INTOL and OUTTOL (iii) TANTO and GOBLACK	06	3	2
	b	Write a manual part program for the component shown below.  Fig. 4b	10	3	2
5	a	Explain the concept of Low-Cost Automation (LCA) and its significance in small and medium enterprises (SMEs).	08	2	2
	b	Imagine you are managing a small manufacturing unit. How would you implement Low-Cost Automation to improve productivity without incurring high expenses? Provide specific example.	08	4	3
		OR			
6	a	A factory is considering the shift from manual assembly to automated assembly lines. Discuss the issues they might face during this transition and how these challenges can be mitigated.	08	4	4
	b	How would you design a strategy for automation in a factory that currently relies heavily on manual labor? Outline the key steps involved.	08	3	3
7	a	Briefly explain the various components of an industrial robot. State the main function of each of the components.	08	2	3
	b	What do you understand by degree of freedom (DOF)? How many DOFs are required to position an end effector at any point in 3-D space?	08	2	2
		OR			
8	a	Explain SCARA robot indicating its joints and degree of freedom with the applications.	08	2	3
	b	What is a robotic joint? Distinguish between prismatic and revolute joints.	08	2	3
9	a	A robotic system in a warehouse is required to sort packages of varying sizes and weights. Discuss how you would select grippers and what programming approach would be most suitable for this application.	08	3	3
	b	You are tasked with programming a robot to perform welding tasks in an automotive assembly line. Describe the programming techniques you would use and justify your choice.	08	3	4

		OR			
10	a	In a scenario where a robot needs to perform a complex assembly task involving multiple components, how would you approach the programming and integration of the robot's sensors and end effectors?	08	4	4
	b	Explain the working principle of an end effector with the help of a sketch and give its important applications.	08	2	3